

KANAGHA SIVAKUMAR

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 github.com/kanaghasivakumar  kanaghasivakumar.github.io/kanagha-portfolio

EDUCATION

Northwestern University

Sep. 2025 – Dec. 2026 (Expected)

M.S. in Artificial Intelligence GPA: 3.97/4.00

Evanston, IL

Teaching (Fall 2026): Graduate Assistant, MSAI 495; Peer Mentor, CS 110

Vellore Institute of Technology, Chennai

Jul. 2019 – May 2023

B.Tech in Computer Science, specialization in AI & ML GPA: 3.71/4.00

Chennai, India

EXPERIENCE

PedzSTAR Lab, Northwestern University | Evanston, IL

Mar. 2026 – Present

Graduate Student Researcher

Child Speech Extraction & Speaker Diarization (3-mic clinical audio)

- Raised adult-speech removal from 47.5% to 72.7% at 97% child-speech recall by fusing a 10 ms Whisper-based child speech detector with 56 cross-microphone crosstalk features.
- Suppressed adult-only speech by 37–95 dB with 0.00 dB median change to child audio by engineering non-generative voice isolation via cross-mic leakage unmixing.
- Deployed on 220 hours of audio (1,529 sessions, 305 children) via SLURM by designing a lossless archive with versioned confidence tracks, enabling full-corpus re-annotation without data re-transfer.

Pediatric Speech Intelligibility Screening

- Flagged high-risk speech pathology at 69.4% accuracy by LoRA fine-tuning 16 ASR models (Whisper, Wav2Vec2, HuBERT) and deriving uncertainty metrics correlated with clinical scores.
- Scaled inference to 131,000+ child utterances with batched GPU pipelines on HPC.

Global Poverty Research Lab (Devidence), Northwestern University | Evanston, IL Jun. 2026 – Jul 2026

Machine Learning Intern

- Built an AI pipeline that pulls key results out of research-paper PDFs using GPT-4o and table detection, tested against 12,000+ expert-entered records from 53 studies.
- Found the pre-existing pipeline's main weak point was locating the right data (387 rows found vs. 143 real) and built a stricter test showing true performance was half the original estimate.

Sunware Technologies Pvt. Ltd. | Chennai, India

Nov. 2023 – Sep. 2025

Associate Software Engineer

Jan. 2025 – Sep. 2025

- Cut manual shipping-data entry by 85%+ for a logistics client by building an LLM OCR pipeline on Gemini 2.0 Flash.
- Mentored interns through a 12-week program; sat on interview panels and recommended interns for full-time roles.

Software Engineer Trainee

Jan. 2024 – Jan. 2025

- Digitized handwritten pilot logbooks at 91% accuracy by pairing AWS Textract and GPT-4 with an aviation dictionary.
- Cut client image-processing time by 90%+ by building an image editor with exposure merging and interactive tuning.

Machine Learning Intern

Nov. 2023 – Jan. 2024

- Forecast pollutant threshold breaches every 15 minutes for Bangalore Police by designing an autoencoder model.

PROJECTS

Speech Synthesis | PyTorch, FastSpeech 2, HiFi-GAN, MFA

2026

- Produced artifact-free 24 kHz speech by building FastSpeech 2 from scratch and fine-tuning HiFi-GAN; verified learned phoneme boundaries via attention-heatmap analysis.

Structured Urban Synthesis | PyTorch, Diffusion Transformers, W&B, SLURM

2026

- Achieved 251.8 FID on Cityscapes with a Diffusion Transformer using cross-attention mask conditioning and CFG.
- Reached 0.013 validation loss on 22,973 images by running W&B Hyperband sweeps across A100 GPUs.

Monodepth Estimation | PyTorch, ResNet, Computer Vision

2026

- Achieved 83.7% $\delta < 1.25$ and 0.137 AbsRel on KITTI via self-supervised ResNet-18 depth + SE(3) pose training.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript (React, Node.js), HTML/CSS **Cloud:** AWS, Azure

ML Frameworks: PyTorch, TensorFlow, Hugging Face (Transformers, PEFT), OpenCV, Scikit-learn, Pandas

Infrastructure: Git, Docker, SLURM/HPC, Weights & Biases, FastAPI, Vector Databases, Globus

Domains: Speech (ASR, Diarization, VAD, TTS), Computer Vision, LLMs (Document AI, Evals), Diffusion, RAG